RADDI-514 Image Processing & Analysis in Diagnostic Imaging

Project Proposal Presentation

University of Alberta

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Improved Segmentation in Thyroid Ultrasound Images Using Combination of Conventional and Deep Learning Techniques

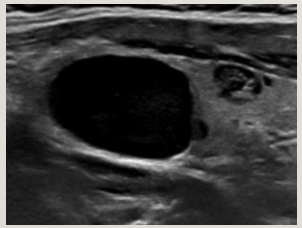
- Thyroid glands – 90% of nodules are benign
- Ultrasound - sensitive imaging modality
- Detection of nodules – Composition , Shape features
- Fine Needle Aspiration – Non-surgical procedure
- Accurate diagnosis – Reduce unnecessary biopsies

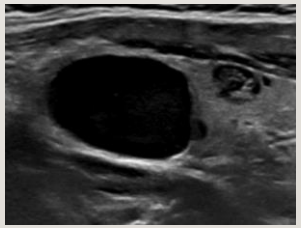
Background

- Visual interpretation – Highly subjective
- Image segmentation
  - Clinically significant
  - Challenging
- Computer Aided Diagnosis– Nodule segmentation and detection

Objective

- User interaction – Single-click approach
- Random points inside nodules for initializing active contour – Independent of mask boundary





Dataset

- Images – Thyroid composition features
- ROI – Thyroid nodules

Thyroid ultrasound images with ground truth
Total: 538

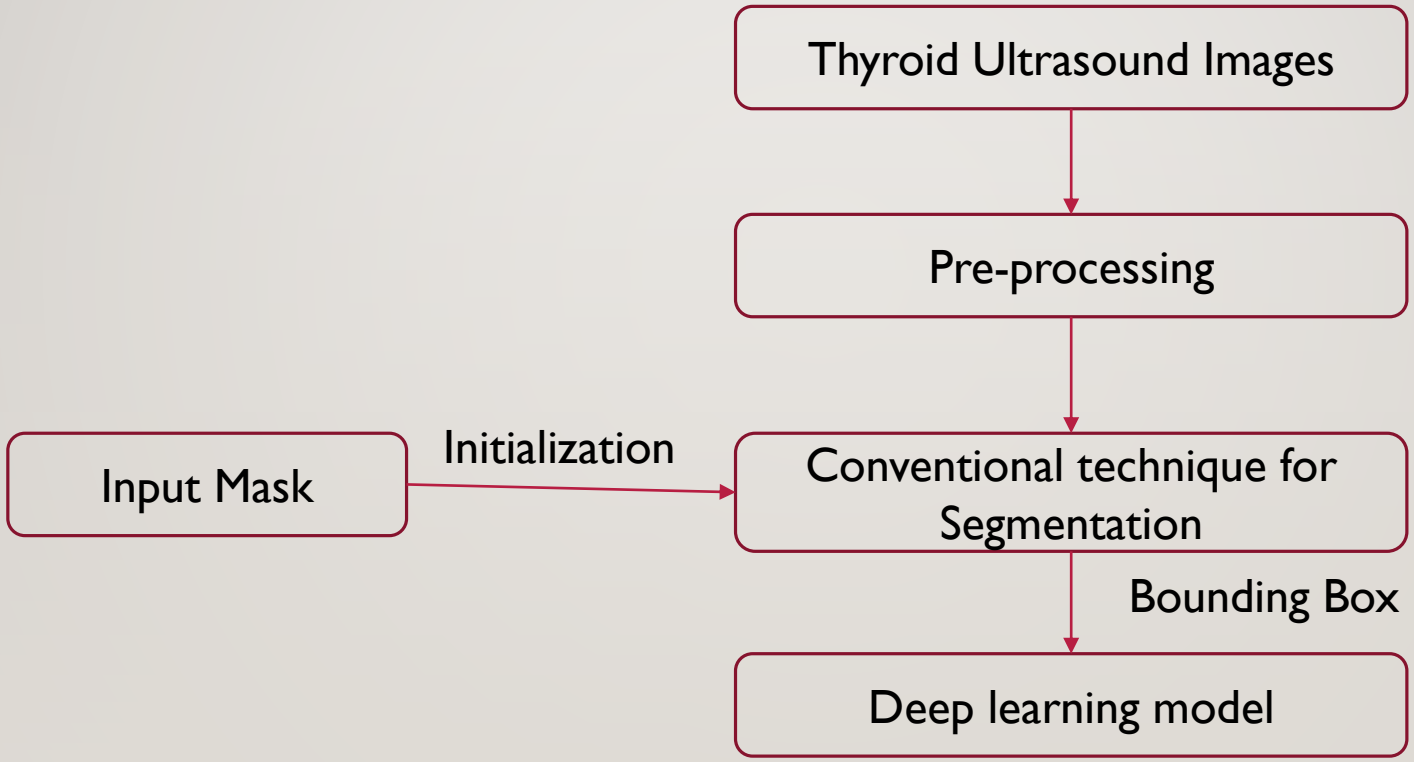
Mixed cystic or solid = 105

Spongiform = 29

Solid or almost completely solid = 347

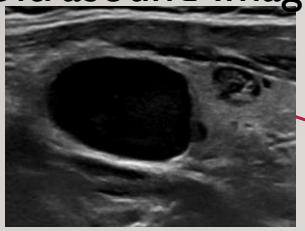
Cystic or almost completely cystic = 57

Method


Thyroid Ultrasound Images
Pre-processing
Input Mask
Initialization
Conventional technique for Segmentation
Bounding Box
Deep learning model

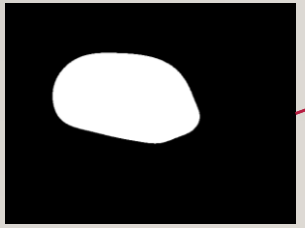
Working

Ultrasound Image



Pre-processing

Input mask



Random points-centre for initialization circle

Active Contour Segmentation

Bounding box

Deep learning model

Improved segmentation

References

- Koundal D et al., "Computer aided thyroid nodule detection system using medical ultrasound images", Elsevier Journal on Biomedical Signal Processing and Control, vol.40, pp.117-130, 2018.
- Hanung et al., "Thyroid nodule segmentation using active contour bilateral filtering on ultrasound images", IEEE Conference on Quality in Research, 2015.
- Ahmet et al., "Comparative MR image analysis for thyroid nodule detection and quantification", Elsevier Journal on Measurement, vol.47, pp.861-868,2014.

Thank You